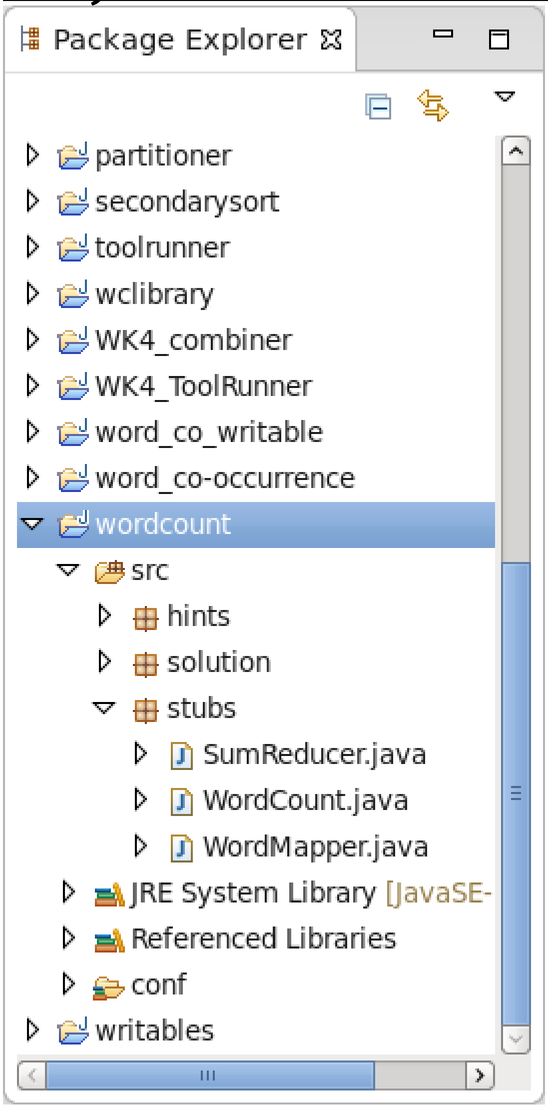
Lab Task 1 (Pass-Level)
Write a MapReduce Program for Finding the Shifted Word Counts

COS20028 – Big data Architecture and Applications

Semester 2, 2022

Name: Put_Your_Name_Here
Student ID: Put_Your_StudentID_Here
Date: Put_the_Submission_Date_Here

Project Folder Structure Screenshot



Driver Code

WordCount.java | WordMapper.java | SumReducer.java

```java
package stubs;

import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
import org.apache.hadoop.mapreduce.Job;

/*
 * MapReduce jobs are typically implemented by using a driver class.
 * The purpose of a driver class is to set up the configuration for the
 * MapReduce job and to run the job.
 * Typical requirements for a driver class include configuring the input
 * and output data formats, configuring the map and reduce classes,
 * and specifying intermediate data formats.
 *
 * The following is the code for the driver class:
 */
public class WordCount {

  public static void main(String[] args) throws Exception {

    /*
     * The expected command-line arguments are the paths containing
     * input and output data. Terminate the job if the number of
     * command-line arguments is not exactly 2.
     */
    if (args.length != 2) {
      System.out.printf(
          "Usage: WordCount <input dir> <output dir>\n");
      System.exit(-1);
    }

    /*
     * Instantiate a Job object for your job's configuration.
     */
    Job job = new Job();

    /*
     * Specify the jar file that contains your driver, mapper, and reducer.
     * Hadoop will transfer this jar file to nodes in your cluster running
     * mapper and reducer tasks.
     */
    job.setJarByClass(WordCount.class);

    /*
     * Specify an easily-decipherable name for the job.
     * This job name will appear in reports and logs.
     */
    job.setJobName("Word Count");


    /*
     * Specify the paths to the input and output data based on the
     * command-line arguments.
     */
    FileInputFormat.setInputPaths(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    /*
     * Specify the mapper and reducer classes.
     */
    job.setMapperClass(WordMapper.class);
```

```java
    job.setReducerClass(SumReducer.class);

    System.out.print("Programmer: Pei-Wei Tsai\nStudent ID: 92144979\n");


    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    /*
     * Start the MapReduce job and wait for it to finish.
     * If it finishes successfully, return 0. If not, return 1.
     */
    boolean success = job.waitForCompletion(true);
    System.exit(success ? 0 : 1);
  }
}
```

Mapper Code

WordMapper.java SumReducer.java

```java
package stubs;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class WordMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

  @Override
  public void map(LongWritable key, Text value, Context context)
      throws IOException, InterruptedException {

    String line = value.toString();

    for (String word : line.split("\\W+")) {
      if (word.length() > 0) {
        context.write(new Text(word), new IntWritable(1));
      }
    }
  }
}
```

Reducer Code

WordMapper.java | SumReducer.java

```java
package stubs;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class SumReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

  @Override
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
            throws IOException, InterruptedException {
        int wordCount = 0;

        for (IntWritable value : values) {

            wordCount += value.get();
        }

        context.write(key, new IntWritable(wordCount));
    }
}
```

Execution Result

training@localhost:~/workspace/wordcount/src

File Edit View Search Terminal Help

```
[training@localhost src]$ pwd
/home/training/workspace/wordcount/src
[training@localhost src]$ javac -classpath `hadoop classpath` stubs/*.java
[training@localhost src]$ jar cvf wc.jar stubs/*.class
added manifest
adding: stubs/SumReducer.class(in = 1592) (out= 666)(deflated 58%)
adding: stubs/WordCount.class(in = 1625) (out= 905)(deflated 44%)
adding: stubs/WordMapper.class(in = 1705) (out= 711)(deflated 58%)
[training@localhost src]$ hadoop jar wc.jar stubs.WordCount shakespeare.tar.gz w
c_res
Programmer: Pei-Wei Tsai
Student ID: 92144979
22/07/25 13:18:07 WARN mapred.JobClient: Use GenericOptionsParser for parsing th
e arguments. Applications should implement Tool for the same.
22/07/25 13:18:07 INFO input.FileInputFormat: Total input paths to process : 1
22/07/25 13:18:07 INFO mapred.JobClient: Running job: job_202207251045_0011
22/07/25 13:18:08 INFO mapred.JobClient:  map 0% reduce 0%
```

Commands for Retrieving and Displaying Results

```
training@localhost:~/workspace/wordcount
File  Edit  View  Search  Terminal  Help
22/07/25 13:18:19 INFO mapred.JobClient:     Total time spent by all reduces waiting after reserving slots (ms)=0
22/07/25 13:18:19 INFO mapred.JobClient:   Map-Reduce Framework
22/07/25 13:18:19 INFO mapred.JobClient:     Map input records=175559
22/07/25 13:18:19 INFO mapred.JobClient:     Map output records=974161
22/07/25 13:18:19 INFO mapred.JobClient:     Map output bytes=8881482
22/07/25 13:18:19 INFO mapred.JobClient:     Input split bytes=117
22/07/25 13:18:19 INFO mapred.JobClient:     Combine input records=0
22/07/25 13:18:19 INFO mapred.JobClient:     Combine output records=0
22/07/25 13:18:19 INFO mapred.JobClient:     Reduce input groups=31835
22/07/25 13:18:19 INFO mapred.JobClient:     Reduce shuffle bytes=10829810
22/07/25 13:18:19 INFO mapred.JobClient:     Reduce input records=974161
22/07/25 13:18:19 INFO mapred.JobClient:     Reduce output records=31835
22/07/25 13:18:19 INFO mapred.JobClient:     Spilled Records=2922483
22/07/25 13:18:19 INFO mapred.JobClient:     CPU time spent (ms)=3140
22/07/25 13:18:19 INFO mapred.JobClient:     Physical memory (bytes) snapshot=315854848
22/07/25 13:18:19 INFO mapred.JobClient:     Virtual memory (bytes) snapshot=1456152576
22/07/25 13:18:19 INFO mapred.JobClient:     Total committed heap usage (bytes)=222167040
[training@localhost src]$ cd ..
[training@localhost wordcount]$ hdfs dfs -get wc_res
[training@localhost wordcount]$ less wc_res/part-r-00000
```

Screenshot of the First Page Result Content